

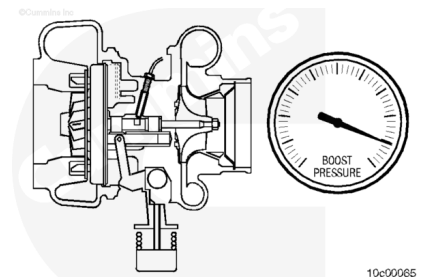
Trainings ▾

General Information

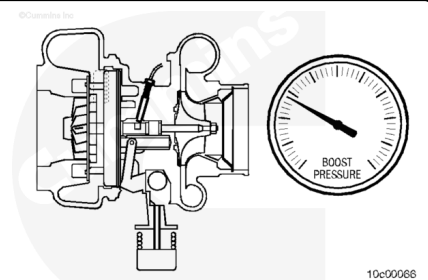
Automotive with CM875, CM870 and CM570

Variable geometry turbocharger purpose:

- When the exhaust gas recirculation (EGR) flow is commanded, the variable geometry turbocharger closes the nozzle in the turbine housing, creating more back pressure in the exhaust manifold to force exhaust gas back into the engine.

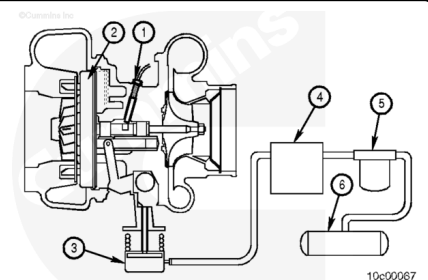


- The turbocharger also functions to improve engine performance by building boost more quickly during acceleration.



The variable geometry turbocharger functions as a standard turbocharger, with the addition of the following:

- A speed sensor (1) in the bearing housing to monitor turbocharger operation
- Water-cooled bearing housings (in addition to oil lubrication)
- A sliding nozzle (2) is actuated by a pneumatic actuator attached to the vehicle (brake) air supply system
- A pneumatic actuator (3) is operated by an air control valve (4)
- On some engines, a filter and shutoff valve (5) are used in the air line between the control valve (4) and the air supply tank (6).



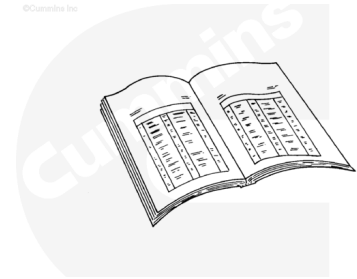
Note : A noise can be heard as air is released from the actuator (3), through the control valve (4), when the variable geometry turbocharger mechanism opens.

Preparatory Steps

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Steam clean the area around the turbocharger actuator and dry with compressed air.
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)

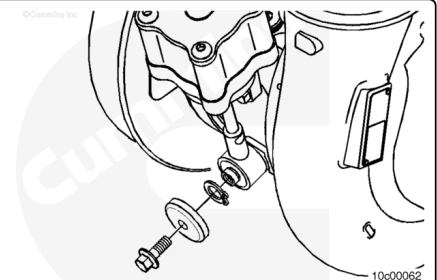
Remove

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⚠ CAUTION ⚠

The zinc anode is a sacrificial washer that reduces the possibility of corrosion of the turbocharger cross-shaft and bushing. The failure to install will result in excessive corrosion and wear to the cross-shaft bushing.

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⚠ CAUTION ⚠

The actuator maintains a spring tension on the variable geometry cross-shaft and can not be removed without releasing this tension by applying air pressure. Damage to the turbocharger can possibly result if air pressure is not applied before removing the actuator.

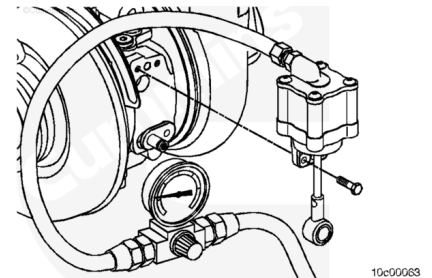
Remove the air line from the actuator if INSITE™ electronic service tool will **not** be used in the removal of the actuator.

Remove the capscrew and zinc anode washer from the turbocharger cross-shaft.

Remove the snap ring from the cross-shaft.

⚠ WARNING ⚠

Keep fingers and hands away from the actuator link to reduce the possibility of personal injury as a result of sudden movement when air is supplied.



Use INSITE™ electronic service tool to override the turbocharger actuator position or apply 620 kPa [90 psi] to 827 kPa [120 psi] of regulated air pressure to the turbocharger actuator.

Remove the two capscrews holding the turbocharger actuator to the turbocharger bearing housing.

The turbocharger actuator end link to cross-shaft is a tight clearance fit. Use care to reduce the possibility of damaging the turbocharger actuator or the turbocharger cross-shaft when removing the actuator.

Remove the turbocharger actuator from the turbocharger.

Turn off the override in INSITE™ electronic service tool or remove the regulated air supply.

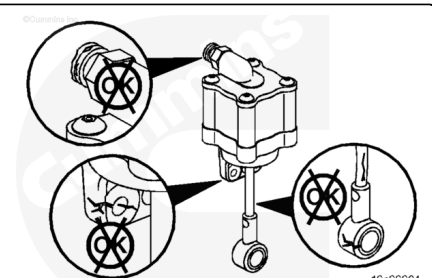
Remove the air line from the turbocharger actuator, if **not** already removed.

Clean and Inspect for Reuse

Automotive with CM875, CM870 and CM570

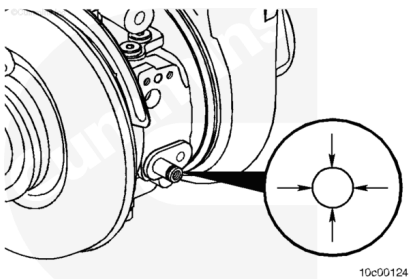
Inspect the actuator mounting bracket, rod, and body. If the actuator is bent or cracked, it **must** be replaced.

Inspect the bushing in the actuator rod end that attaches to the turbocharger cross-shaft for wear, scoring, or other damage. Replace the actuator if damage is found.



Inspect the outside diameter of the turbocharger cross-shaft pin. Replace the turbocharger if wear is found.

Cross-shaft Pin Minimum Diameter		
mm		in
15.1	MIN	0.594



Test

⚠ CAUTION ⚠

There are no serviceable parts inside the actuator, do not disassemble the actuator body.

This test can be performed with the actuator removed or installed on the turbocharger.

Attach a dial indicator as shown, so the shaft is in line with the actuator rod.

Set the dial indicator to zero, with no air pressure applied to the actuator.

Connect clean and regulated pressurized air and a pressure gauge to the actuator. Apply a minimum of 414 kPa [60 psi] to make sure the actuator is functioning properly.

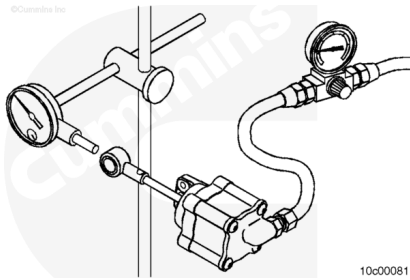
The rod **must** move without sticking.

Actuator Movement Range		
mm		in
12	MIN	0.472

Air **must not** be heard leaking through a functional actuator.

Spray soapy water on the actuator housing to check for air leaks. Replace the actuator housing if leaks are found.

Replace the actuator if no movement of the actuator rod is detected, the actuator is sticking, or an air leak is found.

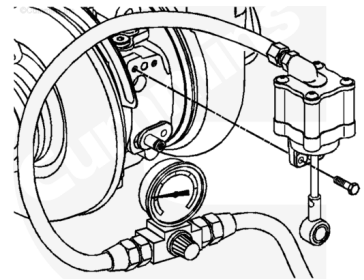


Install

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

Keep fingers and hands away from the actuator link to reduce the possibility of personal injury as a result of sudden movement when air is supplied.



If using INSITE™ electronic service tool to override the turbocharger actuator position, connect the vehicle air supply line to the turbocharger actuator.

If INSITE™ electronic service tool is **not** being used to remove the actuator, apply 620 kPa [90 psi] to 827 kPa [120 psi] of regulated air pressure to the turbocharger actuator.

The turbocharger actuator end link to cross-shaft is a tight clearance fit. Use care to reduce the possibility of damaging the turbocharger actuator or the turbocharger or the turbocharger cross-shaft when removing the actuator.

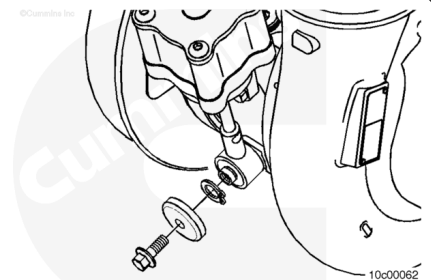
Install the turbocharger actuator onto the turbocharger.

If **not** installing a new turbocharger actuator, apply Threadlocker, Part Number 3824040, onto the two capscrews holding the turbocharger actuator before installing the capscrews onto the turbocharger bearing housing.

Torque Value: 17 n•m [150 in-lb]

⚠ CAUTION ⚠

The zinc anode is a sacrificial washer that reduces the possibility of corrosion of the turbocharger cross-shaft and bushing. The failure to install will result in excessive corrosion and wear to the cross-shaft bushing.



Install the snap ring onto the turbocharger cross-shaft.

Apply Threadlocker, Part Number 3824040, onto the capscrew holding the zinc anode washer before installing the zinc anode washer and the capscrew.

Torque Value: 23 n•m [204 in-lb]

Turn off the override in INSITE™ electronic service tool or remove the regulated air supply.

If **not** already installed, install the air line to the turbocharger actuator.

Start and operate the engine.

Verify proper operation and check for air leaks.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine and check for leaks.

